Unsupervised Prototype-Wise Contrastive Learning for Domain Adaptive Semantic Segmentation in Remote Sensing Image

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***Abstract*— Labeling data in the field of remote sensing is time-consuming and labor-intensive, making domain adaptation between different domains an urgently needed solution. To address the domain gap between diverse datasets in the remote-sensing domain, numerous methods tailored for domain adaptation in high-resolution remote-sensing imagery (RSI) have emerged. Some of the existing methods focus on reducing the domain gap at either the feature level or the pixel level, often overlooking their underlying connection. To tackle this issue, we introduce a prototype-wise contrastive feature alignment (PCFA) paradigm aimed at bridging the representations between the feature and pixel levels. By dynamically updating, we acquire prototype information encompassed by different mini-batches and employ an optimal transport mechanism to reasonably apply the prototype feature distribution in guiding the learning of target domain features. We conduct extensive domain adaptation semantic segmentation (DASS) experiments on the ISPRS Vaihingen and Potsdam datasets, achieving an improvement of about 4%–5% in mean intersection over union (mIoU) compared to previous methods using the DeepLabV2 framework.**

***Index Terms*— Contrastive feature alignment, domain adaptation semantic segmentation, domain gap, feature distribution, optimal transport mechanism, prototype.**

I. Introduction

THE continuous advancement of remote-sensing technology and machinery has continuously improved the efficiency of acquiring high-resolution remote-sensing imagery (RSI). Nowadays, the RSI data is becoming increasingly accepted within the Earth observation community and various application domains. Among these, the task of semantic segmentation of RSI is of paramount importance. It transforms RSI into masks to highlight different regions of interest, with each RSI pixel assigned a unique class label [10]. Semantic segmentation finds applications in various domains, including land use, change detection, land cover, and urban planning [15], [20].

Semantic segmentation methods for RSI have benefited from the vigorous development of supervised deep-learning techniques [4], [30], [42], [48], [49], [50], [51]. For instance, methods like deep convolutional neural networks and their variants have achieved significant performance improvements when applied to RSI semantic segmentation [38], [39], [40]. These approaches primarily rely on supervised deep-learning methods to train models, which in turn heavily depend on a large number of manually annotated images. Manual annotation of images is a time-consuming and labor-intensive task.

To reduce the cost of manual annotation, semantic segmentation methods for RSI based on weakly supervised and unsupervised techniques are gradually emerging. Unsupervised domain adaptation semantic segmentation (DASS) strategies, as a promising approach, has received significant research attention and achieved notable performance improvements [14]. The objective of unsupervised DASS methods is to transfer segmentation knowledge from densely labeled source domains to unlabeled target domains. For instance, a category-based generative adversarial network has been developed to address intra- and interclass disparities in unsupervised DASS semantic segmentation for RSI [22]. Considering the disparities among different classes and regions [1], a local-to-global unsupervised DASS segmentation framework has been proposed. Most of these studies primarily employ single-modal data, specifically orthoimages, for model training. However, orthoimages contain relatively uniform information, which imposes certain limitations. For example, in regions where ground objects share similar spectral characteristics or are situated in shadowed areas, segmentation models struggle to differentiate effectively, often leading to misclassification between objects such as buildings and roads, or low vegetation and trees. Some classical methods from the natural domain, such as self-training (ST), have introduced a teacher–student network. This approach guides the learning of the student network by generating pseudo-labels using the teacher network, thereby bridging the gap between the two domains. Additionally, some methods align the source and target domains at either the feature level [17] or the pixel level [18], offering new insights for unsupervised DASS methods in the RSI domain.

Manuscript received 19 September 2023; revised 26 October 2023; accepted 14 November 2023. Date of publication 17 November 2023; date of current version 30 November 2023. This work was supported in part by the Key Scientific Technological Innovation Research Project by the Ministry of Education; in part by the National Natural Science Foundation of China under Grant 62171347, Grant 61877066, Grant 62276199, Grant 61771379, Grant 62001355, and Grant 62101405; in part by the Science and Technology Program in Xi'an of China under Grant XA2020-RGZNTJ-0021; and in part by the 111 Project. *(Corresponding author: Biao Hou.)*

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Digital Object Identifier 10.1109/TGRS.2023.3334294

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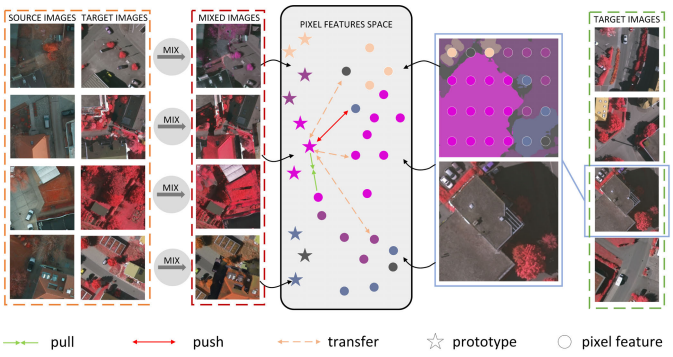


Fig. 1. Prototype by bringing positive samples closer together and pushing away negative samples during contrastive learning. It uses the optimal transport theory to guide the network in learning the pixel feature distribution in the target domain.

However, due to the inherent complexity of RSI segmentation tasks, the results have not been entirely satisfactory.

To address the aforementioned issues, this article introduces a prototype, serving as a bridge to mitigate the gap between two domains, not limited to aligning differences solely at the feature or pixel level. In addition, it leverages contrastive learning to bring positive samples closer and negative samples farther apart in both feature and pixel domains. To endow the prototype with memory during the continuous network updates, preserving historical information, a dynamic update module is devised in this work. During the learning process of target domain images, to further reduce the gap between the two domains, this article proposes a novel optimal transport loss based on feature distribution space, guided by prototypes that incorporate information from both the source and target domains. We summarize the role of our prototype in Fig. 1, and we outline our contributions as follows.

1) We propose an unsupervised DASS contrastive learning paradigm that aligns the source and target domains at both the feature and pixel levels.
2) We introduce a prototype as a bridge between the source and target domains and devise a novel dynamic update module to intelligently update the prototype.
3) We formulate a novel loss leveraging optimal transport theory to optimize the feature distribution in the target domain using the prototype.
4) Extensive experiments were conducted on the Vaihingen and Potsdam datasets, resulting in significant achievements.

II. Related Works

A. Unsupervised DASS in Natural Images

Unsupervised DASS [44], [45], [46] has found success in the realm of natural images. As domain alignment techniques have advanced in image classification tasks, various alignment strategies [7], [23], [25], [28] have also been successfully applied in the field of semantic segmentation. Particularly for DASS, some pipelines have delved into optimizing features in different spatial domains. For instance, [13] focused on minimizing the alignment of boundary feature distributions between the source and target domains at the feature level. Furthermore, [27] established pixel-level alignment methods by constructing different style transfers to establish connections between domains. With the development of semisupervised learning, the generation of pseudo-labels has brought forth many new approaches. A classic self-supervised learning paradigm, ST [19], constructs teacher networks to generate pseudo-labels for student network learning, reducing the gap between domains and offering new perspectives for subsequent DASS task designs. However, unsupervised tasks not only rely on providing pseudo-labels for guidance, but also introduce the concept of contrastive learning, offering hope for distinguishing relatively indistinguishable features in semisupervised settings. Inspired by contrastive learning, CLST [26] constructs prototypes through interclass pixel generation, utilizing prototypes to bring positives closer and push negatives farther away. ProCA [6] directly contrasts pixel representations with prototypes, achieving domain alignment at the pixel level. These methods have inspired unsupervised DASS in our remote-sensing domain, with few attempts to integrate these ideas into a unified optimization framework.

B. Unsupervised DASS in Remote-Sensing Images

In the field of remote sensing, the cost of high-quality annotation has become a significant issue. To minimize dependence on high-quality labels, unsupervised DASS methods have emerged. These methods can be categorized into three types: image-to-image translation, adversarial learning, and self-supervised domain adaptation. Image-to-image translation first transforms the colors and textures of the source domain to resemble those of the target domain and then trains a segmentation model on the source domain [34]. As the visual gap between the source and target domains diminishes, the model becomes more adaptable to the target domain. For example, CycleGAN [11] and Fourier domain adaptation (FDA) [24] exchange low-frequency spectra using Fourier transformation and its inverse. Unfortunately, during mapping, translated images may introduce considerable noise, leading to annotation mismatches between the original source-domain images and the translated source-domain images. Research in adversarial learning has seen continuous development. Hoffman et al. [2] first applied adversarial training to unsupervised domain adaptation (UDA) for semantic segmentation. To consider cross-domain structural consistency, ADVENT [12] further optimized the distribution alignment performance in the output space using weighted self-information. However, these methods did not account for interclass independence and differences between different regions in images. To address both of these issues, class-level adversarial networks (CLANs) [18] were proposed, introducing CLANs to ensure local semantic consistency during global alignment. The core idea of self-supervised learning is to train a model by generating pseudo-labels from the model's previous states. Selective pseudo-labeling (SPL) [9] and robust class matching (RCM) [16] generate pseudo-labels by mapping images from the source and target domains into a unified feature space to achieve class-wise feature alignment. However, when applied

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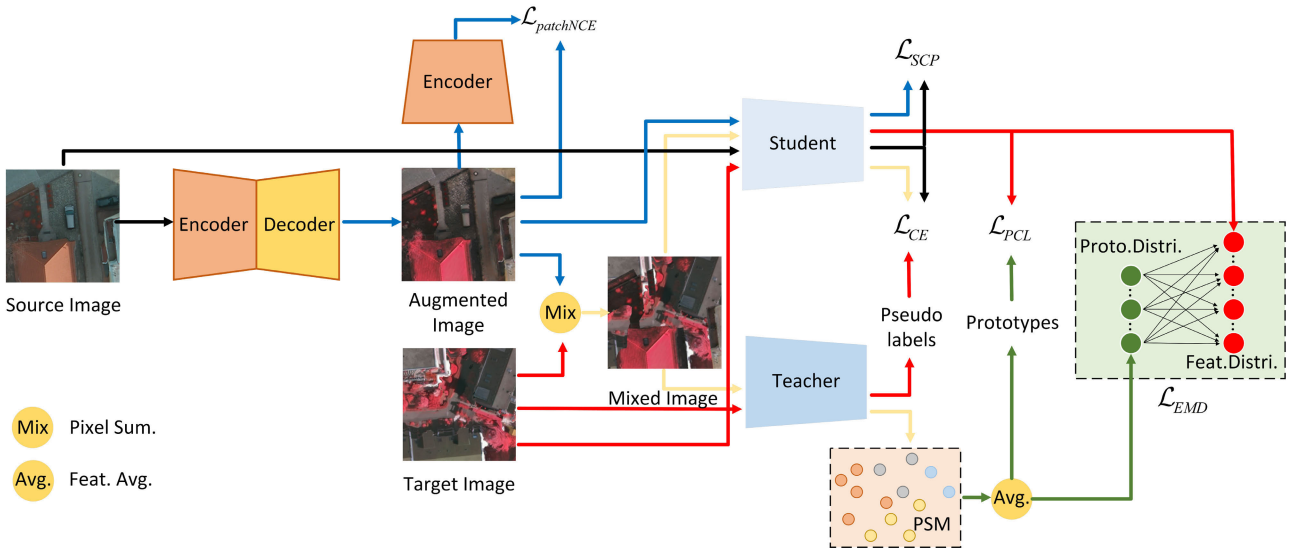


Fig. 2. Our work's overall workflow framework. The source domain images are initially processed through a stylization network, transforming them into enhanced images. These enhanced images are then fused with images from the target domain and subsequently fed into a teacher–student network to perform domain adaptation tasks. Ultimately, through the collaborative influence of PSM and prototype-guided feature distribution, a notable improvement in performance is achieved.

to semantic segmentation, both approaches treat each pixel as a separate representation, overlooking crucial contextual relationships. DAST [35] assigns higher weights to challenging regions using an attention mechanism and aligns both the feature space and the output space. Faced with the problem of class imbalance in segmentation, RCA-DD [33] increases training samples by transferring content from tail classes in the source domain to the target domain.

In this article, we leverage the concept of contrastive learning and introduce it into the foundation of self-supervised learning. We employ a prototype-wise framework and perform domain alignment at both the feature and pixel levels to reduce the gap between domains. Moreover, we utilize optimal transport theory to optimize the learning of the feature distribution in the target domain using prototypes. The framework is depicted in Fig. 2, and we will provide a detailed analysis in Section III.

III. Method

In this work, to better explore the latent connections between different datasets in domain adaptation, we introduce prototype-wise contrastive feature alignment (PCFA), an UDA semantic segmentation algorithm. By learning prototypes that capture common features between domains and leveraging the principles of contrastive learning, we bring pixels of the same class in the target domain closer while pushing the features of different class samples farther apart. Before delving into the main contributions of this article, we will first analyze the paradigms in domain adaptation and discuss some fundamental concepts involved in the domain adaptation domain.

A. Preface

In DASS, we have access to a labeled source-domain dataset $D_s = \{I_1^s, \ldots, I_{N_s}^s, L_1^s, \ldots, L_{N_s}^s\}$ and an unlabeled target-domain dataset $D_t = \{I_1^t, \ldots, I_{N_t}^t\}$. Here, $I_i^{s,t}$ represents the images from the source and target domains, while L_i^s represents the labeling information present in the source domain. Specifically, this information takes the form of a one-hot pixel-level segmentation mask $L_i^s \in \mathcal{R}^{H\times W\times C}$, where C denotes the total number of classes in the dataset. The network architecture primarily consists of a feature extractor $f_{\text{ext}}(x)$ and a segmentation decoder $f_{\text{dec}}(x)$, both working in conjunction to form the network $f(x) = f_{\text{ext}}(x) \odot f_{\text{dec}}(x)$.

1) Self-Supervised Learning: In domain adaptation for semantic segmentation tasks, one common approach is self-supervised learning—an effective unsupervised method that leverages a teacher–student network paradigm [19] for cross-domain segmentation tasks. In this process, the teacher network $f_{\text{tea}}(x)$ is utilized to provide one-hot pseudo-labels for training the student network $f_{\text{stu}}(x)$ on unlabeled target-domain images. Subsequently, the target-domain data with pseudo-labels $\hat{L}_i$ are used for training the student network, typically employing the standard cross-entropy (CE) loss

$$\mathcal{L}_{\text{CE}} = -\frac{1}{N_t}\frac{1}{HW}\sum_{j=1}^{H\times W}\sum_{i=1}^{C}\hat{L}_i \log p_j^i. \tag{1}$$

Here, $p_j^i = f(I_j^t)$ represents the probability that the jth pixel in the student network belongs to the i class. To ensure that the source-domain data also receives appropriate training, we apply this formula to the labeled source-domain data. To reduce the noise introduced by pseudo-labels for the target-domain dataset, we employ a data augmentation technique to enhance data generalization [29].

In the basic ST method [19], to enhance the model's generalization, reduce random fluctuations, and facilitate model convergence, we often employ the exponential moving average (EMA) technique. This involves using the parameters θ of the student network to update the parameters $\tilde{\theta}$ of the teacher

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network, and this process occurs at each iteration $iter$

$$\tilde{\theta}_{\text{iter}+1} = \alpha\tilde{\theta}_{\text{iter}} + (1-\alpha)\theta_{\text{iter}}. \tag{2}$$

Here, α is the parameter for momentum updates, and we adopt the default value of 0.999.

B. Contrastive Pixel-Level Alignment

Inspired by [21], our proposed PCFA introduces the contrastive idea, utilizing the InfoNCE [3] loss to optimize the relationships between feature pairs during network learning. Considering some paradigms proposed in the DASS domain, it is found that pixel-level alignment is an effective and crucial technique. In some aspects, it outperforms feature-level alignment [31], [32]. Therefore, to construct pixel-level positive and negative sample pairs in contrastive learning, we treat pixel features f^s in the source-domain images as samples and use a data augmentation module to generate new augmented versions of pixel features f^{aug} resembling the source domain. For each pair, the pixel features at the corresponding positions serve as positive samples, while the features at other positions are considered negative samples. We summarize this as follows:

$$\begin{aligned}\mathcal{L}_{\text{patch-NCE}} &= -\mathcal{Z}_{f^{\text{aug}}} \log p_{f^{\text{aug}}} \\ p_{f^{\text{aug}}} &= \frac{\exp\left(\frac{f^{\text{aug}} \cdot f^s_+}{T}\right)}{\exp\left(\frac{f^{\text{aug}} \cdot f^s_+}{T}\right) + \sum_{i=1}^{H\times W} \exp\left(\frac{f^{\text{aug}} \cdot f^s_{-,i}}{T}\right)}\end{aligned} \tag{3}$$

where $\mathcal{Z}_{f^{\text{aug}}}$ represents a mean operation, and the numerical value should be $1/(H \times W - 1)$.

C. Prototypes Design

Because standard ST training only considers local information for given pixels and overlooks global information between datasets, relying solely on pixel-level alignment is far from sufficient. Furthermore, in an unsupervised setting, pseudo-labels can easily introduce noise without supervision, thereby affecting the learning of target domain data.

1) Generation of Prototypes: In many recent studies on contrastive learning [36], [37], [41], feature-level alignment is used to adapt contrastive learning between pixels. However, in DASS tasks, establishing a connection between two different domain datasets is crucial. Considering some prototype-based contrastive learning methods [6], [26], [47], which utilize prototypes with more global information to constrain features, pulling features of the same class closer and pushing features of different classes farther apart, we generate prototypes by merging source- and target-domain images, thus using images that contain information from both domains to create prototypes. This enhances the model's ability to learn latent information between different domains. However, using fused images alone as prototypes can introduce a lot of noise and does not provide reasonable constraints for different classes. Considering that class activation maps (CAMs) reflect how the model responds to different class images and indicate the model's attention to learning objectives, we were inspired to use a CAM-weighted approach to derive the prototypes in our work.

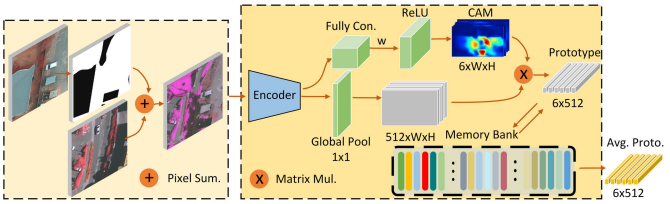


Fig. 3. Explanation of how the prototypes in this article are generated. We utilize generated CAMs as weights and fuse them with network features to obtain prototypes. Building upon this, we design a memory bank to acquire additional historical prototype information during the training process.

2) Calculation of prototypes: The computation of CAM does not impose a significant computational burden; however, it effectively highlights regions of interest that the network deems more important. Although CAM may not strongly emphasize the boundaries of targets, in the DASS task, our primary focus is the pixel-level feature classification performance. Boundary classification errors do not significantly impact our task. By constructing a fully connected layer in the decoder, given input features $f = f_{\text{ext}}(x)$, $f \in \mathcal{R}^{\hat{H}\times\hat{W}\times\hat{D}}$, the features are dimensionally compressed through global average pooling (GAP) and then fed into a linear layer to obtain scores for each class. Here, we define CAM as M_c, with the parameters in the linear layer denoted as $Parms \in \mathcal{R}^{C\times\hat{D}}$

$$M_{\text{c}} = \text{ReLU}\left(\sum_{i}^{\hat{D}} \text{Parms}_{c,i} \cdot f_i\right). \tag{4}$$

Here, we remove all negative values to obtain the CAM for each desired class. On this basis, we use them as weights and multiply them with the images in each mini-batch. We then extract the positions of the top-n highest CAM pixel values, which represent the prototype $P = \{\text{p}_1, \ldots, \text{p}_C\}$

$$\text{p}_c = \sum_{i}^{n} M_{c,i} \cdot \tilde{f}_i \Big/ \sum_{j\neq i}^{n} M_{c,j}. \tag{5}$$

We organize the results as shown in Fig. 3, and since the prototypes are obtained within each mini-batch, to align with our parameter updating approach in ST, we also apply EMA to iteratively update the prototypes

$$\text{p}_c^{\text{iter}+1} = \beta\text{p}_c^{\text{iter}} + (1-\beta)\text{p}_c^{\text{iter}}. \tag{6}$$

Here, p_c^{iter} represents the prototype obtained within the current mini-batch, and β is the momentum update parameter, which is also set to 0.999.

3) Prototype Storage Module: The update of prototypes is primarily determined by the pixel features within the current mini-batch. However, we aim to enable the network to focus on more information contained in pixel values simultaneously. Drawing inspiration from the concept of a memory bank in contrastive learning [36], we designed an asynchronously updated prototype storage module (PSM) that includes historical information. By collecting prototypes obtained from different mini-batches and continually incorporating the latest information from the current mini-batch, we ensure that the storage contains the expected amount of prototype features. This allows us to surpass the limitations of batch quantity

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and, within the confines of available memory, ensure that the network learns prototype features with richer information. To leverage all the stored information, we update prototypes as $\mathrm{p}_c = \sum_{i=1}^{N_{psm}} \mathrm{p}_c^i / N_{\mathrm{psm}}$. The structure of this module is illustrated in Fig. 3. In the subsequent experimental section, we conducted ablation experiments to validate the feasibility of the module, and in this work, we set the PSM storage capacity to 250.

D. Contrastive Prototypes Domain Alignment

Prototypes often represent commonalities among the same classes. After obtaining the prototypes required for target-domain learning through the above methods, inspired by contrastive learning, we apply the prototypes in the contrastive optimization of pixel features in this article. Unlike conventional contrastive learning, positive and negative samples are replaced with more representative prototype features, which can guide similar pixels to cluster together and dissimilar pixels to move apart. Moreover, the prototypes used in this article include pixel information from both the source and target domains, establishing a strong bridge between the two domains. To enhance the representativeness of the information encoded in the prototypes, an asynchronous updating mechanism is introduced to expand the potential scope of prototype representation.

We are performing contrastive optimization between all pixel features contained in each mini-batch of images and prototypes. First, we summarize the formula as follows:

$$\mathcal{L}_{\mathrm{PCL}} = -\frac{1}{N}\sum_{i=1}^{N} L_{i,c} \log \frac{\exp(f_i \cdot \mathrm{p}_c / T)}{\sum_{c=1}^{C} \exp(f_i \cdot \mathrm{p}_c / T)}. \tag{7}$$

Here, $L_{i,c}$ represents the real labels in the source domain and pseudo-labels in the target domain, T represents the temperature coefficient, and p_c represents the prototype features of the same category as the current pixel feature f_i.

Due to the variations in color, brightness, texture, and other factors introduced by data augmentation as shown in the figure, inspired by [21], we use the SC [43] loss to reduce the difference between the source-domain images and the augmented images. Considering that prototypes can be used to supervise network learning in this article, we introduce prototypes into the learning of context (source domain and augmented images). The loss function is summarized as follows:

$$\mathcal{L}_{\mathrm{SCP}} = \frac{1}{\hat{H}\hat{W}C}\sum_{i=1}^{\hat{H}\times\hat{W}}\sum_{c=1}^{C}\left\| f_i^s \cdot \mathrm{p}_c - f_i^{\mathrm{aug}} \cdot \mathrm{p}_c \right\|^2. \tag{8}$$

Here, f_i^{aug} represents the pixel features obtained through online parameter updates in the data augmentation module.

E. Feature Distribution Shift

1) Optimal Transport: In our previous sections, we designed losses at both the feature and pixel levels. However, different sets of features have their own distributions. Therefore, in DASS, we aim to establish a connection between the data distribution of target-domain images and the prototype set from a data distribution perspective. Wasserstein distance, also known as Earth Mover's distance (EMD) [8], is a measure of the minimum cost required to transform one distribution into another (the minimum average distance of movement). In this article, the prototypes serve as a bridge between the source and target domains, and we guide the distribution of target-domain data by exploring their underlying representations. This enhances the network's classification performance on different classes within the target domain. Hence, we design the prototype-guided EMD loss, which is summarized as follows:

$$\mathcal{L}_{\mathrm{EMD}} = \inf_{\gamma \sim \prod(P_{\mathrm{p}}, P_t)} \left(\mathrm{E}_{(x,y)\sim\gamma}\left[\|x-y\|\right]\right). \tag{9}$$

$\prod(P_{\mathrm{p}}, P_t)$ represents the set of all possible joint distributions formed by combining the prototype distribution and the target-domain distribution. For each possible joint distribution γ, we can sample from it $(x, y) \sim \gamma$ and compute the distance $\|x - y\|$ to the samples and take the mean $E[\|x - y\|]$. To perform the lower bound operation $\inf(x)$, we take the lower bound of these expected values among all possible joint distributions, which is our EMD loss.

2) Joint Training Objective: Building upon the foundation of ST, we introduce contrastive learning through prototypes and design a novel prototype-optimal transport loss to jointly form the overall loss function in this article

$$\mathcal{L}_{\mathrm{all}} = \underbrace{\mathcal{L}_{\mathrm{CE}}^{S,M}}_{\mathrm{ST}} + \underbrace{\mathcal{L}_{\mathrm{patch-NCE}}^{S}}_{\mathrm{Contrastive\ Learning}} + \underbrace{\mathcal{L}_{\mathrm{PCL}}^{S,M} + \mathcal{L}_{\mathrm{SCP}}^{S}}_{\mathrm{Prototype}} + \underbrace{\mathcal{L}_{\mathrm{EMD}}^{T}}_{\mathrm{Prototype}\rightarrow\mathrm{Target}}. \tag{10}$$

Here, S, M, and T represent the source-domain image, the fused image, and the target-domain image, respectively. We evaluate the joint optimization effect of the above functions in the subsequent experimental section by introducing corresponding losses at different stages for network optimization.

IV. Experiments

A. Dataset Benchmark

ISPRS provides two state-of-the-art airborne image datasets for urban classification and 3-D building reconstruction tasks. These datasets incorporate digital surface models (DSMs) generated using high-resolution orthophotos and corresponding dense image-matching techniques. Both datasets cover urban scenes. Vaihingen represents a relatively small village, with numerous independent buildings and small multistory structures. Potsdam, on the other hand, is a typical historical city characterized by large building complexes, narrow streets, and dense settlement structures. Each dataset has been manually classified into six of the most common land cover classes: "impervious surfaces," "building," "car," "tree," "low vegetation," and "clutter/background."

1) Vaihingen Dataset: Vaihingen[1] comprises 33 remote-sensing images of varying sizes. The spatial resolution of the

[1]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-vaihingen.aspx

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top-level imagery and DSM is 9 cm. Remote-sensing images are in the format of 8-bit TIFF files, consisting of three bands: near-infrared, red, and green. The DSM is a single-band TIFF file with 32-bit floating-point encoding (corresponding to DSM heights). In the experiments, due to GPU memory limitations, we randomly cropped the original dataset into 3000 images of size 512×512 pixels. When the dataset is designated as the source-domain dataset, all images are used for training. When it is set as the target-domain dataset, 30% of the data is allocated for testing, while the remaining 70% constitutes the training set.

2) Potsdam Dataset: Potsdam[2] comprises 38 images of identical size, with fixed dimensions of 6000×6000 pixels. The spatial resolution of the top-level imagery and DSM is 5 cm. Similar to the Vaihingen region, this dataset consists of three-band remote-sensing TIFF files and single-band DSM data. In our experiments, we randomly cropped it into 6000 images of size 512×512 pixels. We use both RGB and IRRG format data in the dataset as representations of two distinct domains. Likewise, when utilized as the source-domain imagery, all data are employed for training. When used as the target-domain dataset, the ratio of the testing set to the training set is 3:7.

B. Implementation Details

We utilized DeepLab-V2 with ResNet-101 as the backbone architecture in our approach. Optimization was performed using the AdamW optimizer, with a learning rate set to 6×10^{-5} and a weight decay of 0.01. Other fundamental settings for domain DASS were aligned with ST.

For the sample collection from the dataset, we employed a rare class sampling strategy and generated mixed images following the guidance from [21]. The parameters for the data augmentation module were primarily based on [29]. The feature dimension of the prototypes was set to 512.

C. Comparison With Other Algorithms

To validate the performance of our unsupervised DASS framework in the remote-sensing domain, we selected the ISPRS Vaihingen and Potsdam datasets for the experimental evaluation of our approach. For the Vaihingen dataset, we chose IRRG-format data, while Potsdam provided both IRRG- and RGB-format images. To assess the transferability of our approach across different types of datasets, we conducted four sets of experiments. These experiments included both transfers between datasets of the same type (IRRG to IRRG) and transfers between datasets of different types (IRRG to RGB). We performed detailed analyses and visualization of the results, which are presented in the following figures for further insights.

1) Vaihingen-IRRG to Potsdam-IRRG Transfer: Domain adaptation between datasets of the same type proved to be effective in the case of Vaihingen-IRRG to Potsdam-IRRG. However, due to the limited quantity of training images from Vaihingen compared to Potsdam, the final performance was 3.17% lower than our results in Potsdam-IRRG to Vaihingen-IRRG. Nevertheless, there was approximately a 9% improvement compared to other comparative methods. As observed in Fig. 4, our approach exhibited finer pixel-level details compared to other methods, some of which suffered from loss or misclassification issues, for instance, ADVENT [12] and CLAN [18]. However, when training with only the SOURCE strategy, our approach might miss detecting certain classes, as seen in the last row where the background class was not detected, and in the first row where some information for the tree class was lost. In Table I, in terms of IoU for various classes, our approach showed significant improvements compared to others, with the exception of the background class. This is because the background class, due to its inherent complexity and limited sample size in the dataset, is challenging to detect and does not show a distinct advantage.

2) Vaihingen-IRRG to Potsdam-RGB Transfer: Performing domain adaptation between datasets of different types, as in the case of Vaihingen-IRRG to Potsdam-RGB, presents increased challenges. Additionally, the larger number of target domain test samples compared to the limited source domain training samples contributed to slightly lower performance metrics, as indicated in Table II. From Fig. 5, it is evident that our approach captured a substantial amount of image information compared to other tasks, but there were still regions with missed detections in the final results. Analyzing IoU scores, some classes in our approach, such as "low vegetation," showed significant disparities compared to ADVENT [12]. However, this variation among classes is expected in such domain-adaptation tasks, where different methods may exhibit advantages in specific categories. Nevertheless, overall, our approach achieved approximately a 9% advantage in mean intersection over union (mIoU), demonstrating its competence in challenging domain-adaptation tasks.

Potsdam-IRRG to Vaihingen-IRRG Transfer. In the domain-adaptation task between datasets of the same type, Potsdam-IRRG to Vaihingen-IRRG, where the training dataset has the maximum number of samples, our approach demonstrated outstanding performance within the set of experiments on unsupervised DASS in this article. There was a remarkable improvement of approximately 12% compared to other algorithms. Furthermore, our approach exhibited significant advantages in terms of IoU for each class. As depicted in Fig. 6, for the last row, which represents a scenario with complex pixel information, it is evident that our approach accurately detected the pixel categories and their respective positions for each class in the image. This demonstrates that our approach can achieve excellent performance in domain-adaptation tasks between datasets of abundant data and the same data type in Table III.

3) Potsdam-RGB to Vaihingen-IRRG Transfer: Under the condition of an equal number of training samples but different data types, our approach still achieved an approximately 5% higher mIoU compared to Vaihingen-IRRG to Potsdam-RGB. As observed in Fig. 7, other comparative methods demonstrate a detailed grasp of the overall image

[2]https://www.isprs.org/education/benchmarks/UrbanSemLab/2d-sem-label-potsdam.aspx

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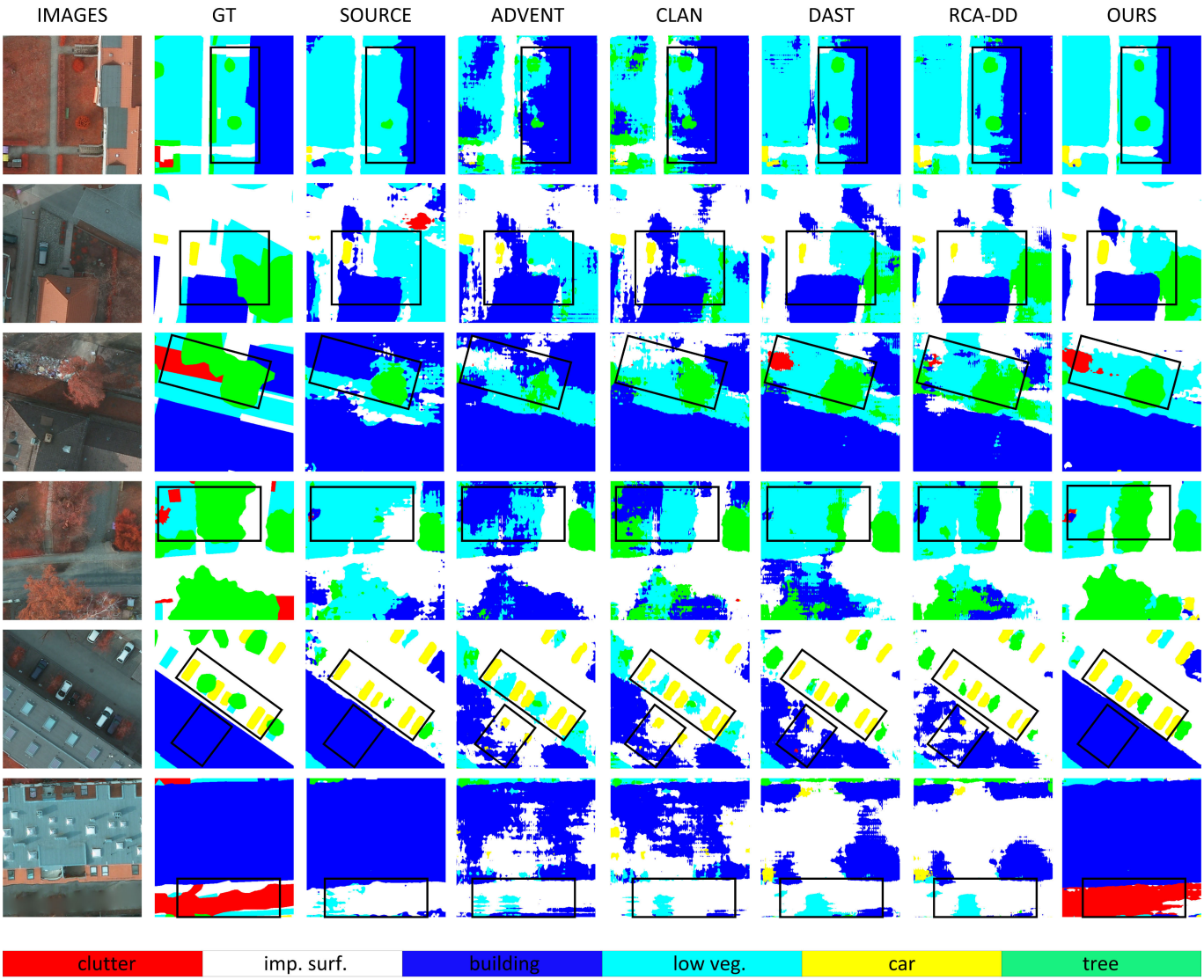


Fig. 4. Visualization results for the unsupervised DASS from Vaihingen IRRG to Potsdam IRRG. The black-bordered sections represent regions for ease of comparison. From left to right are the original image, ground-truth labels, source-only training, ADVENT [12], CLAN [18], DAST [35], RCA-DD [33], and ours (PCFA).

TABLE I

COMPARISON OF UNSUPERVISED DASS RESULTS OF THE VAIHINGEN IRRG TO POTSDAM IRRG DATASET. THE RESULTS ARE IOU (%). (∗) REPRESENTS EXPERIMENTAL RESULTS CITING OTHER ARTICLES. (†) DENOTES A REPLICATION EXPERIMENT USING THE DATASET DIVIDED IN THIS ARTICLE

Methods	Vaihingen IRRG → Potsdam IRRG						
	impervious surfaces	car	tree	low vegetation	building	clutter/background	mIou
†SOURCE ONLY	46.63	55.15	44.31	12.94	62.57	1.49	37.18
*ADVENT [12]	48.40	39.72	27.26	43.56	52.88	12.10	37.32
*CLAN [19]	50.95	39.17	25.50	46.39	53.27	13.21	38.08
*DAST [36]	57.40	41.59	34.32	47.73	57.90	13.07	42.00
*RCA-DD [34]	56.13	45.46	34.43	43.31	61.65	12.74	42.29
*DSM-A [53]	64.75	47.06	34.45	48.39	77.33	13.56	47.59
*CaGAN [56]	56.78	34.57	36.96	37.26	51.21	9.38	37.69
*DecoupleNet [55]	59.34	50.38	42.19	56.98	64.30	10.45	47.27
*MuGCDA [54]	60.12	52.29	43.63	58.15	65.19	15.76	49.19
PCFA(ours)	67.28	89.13	85.99	51.52	91.43	14.89	**51.23**

information. Upon closer examination of the details, our approach exhibits a significant advantage. As shown in Table IV, for the challenging background class segmentation, our approach has an absolute advantage, indicating excellent performance in domain adaptation for such difficult classes.

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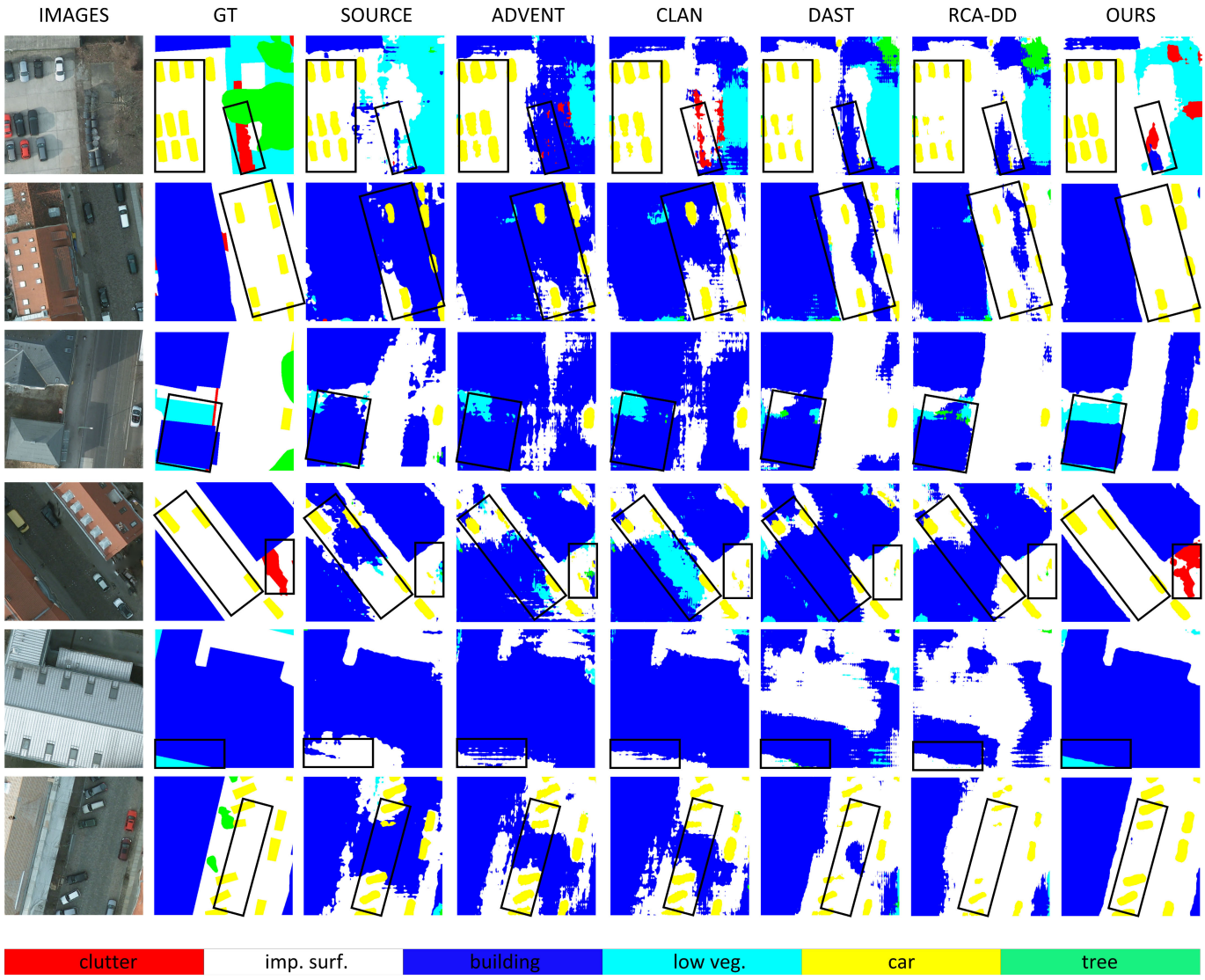


Fig. 5. Visualization results for the unsupervised DASS from the Vaihingen IRRG to Potsdam RGB dataset. The black-bordered sections represent regions for ease of comparison. From left to right are the original image, ground-truth labels, source-only training, ADVENT [12], CLAN [18], DAST [35], RCA-DD [33], and ours (PCFA).

TABLE II

COMPARISON OF UNSUPERVISED DASS RESULTS OF THE VAIHINGEN IRRG TO POTSDAM RGB DATASET. THE RESULTS ARE IOU (%). (∗) REPRESENTS EXPERIMENTAL RESULTS CITING OTHER ARTICLES. (†) DENOTES A REPLICATION EXPERIMENT USING THE DATASET DIVIDED IN THIS ARTICLE

Methods	Vaihingen IRRG → Potsdam RGB						
	impervious surfaces	car	tree	low vegetation	building	clutter/background	mIou
†SOURCE ONLY	32.23	49.73	39.43	5.30	42.35	1.02	28.34
*ADVENT [12]	42.12	37.24	29.85	27.44	49.84	10.24	32.79
*CLAN [19]	45.64	36.97	21.94	26.99	39.42	10.98	30.32
*DAST [36]	49.51	39.23	41.34	6.75	44.76	11.42	32.17
*RCA-DD [34]	49.65	39.68	32.13	12.76	58.39	12.88	34.25
*DSM-A [53]	59.54	49.40	34.73	20.96	68.16	14.39	41.20
*CaGAN [56]	57.76	70.12	42.09	15.11	58.27	6.37	41.62
*DecoupleNet [55]	56.10	63.84	38.29	17.52	58.93	10.21	40.82
*MuGCDA [54]	58.35	65.29	40.28	19.32	59.47	8.49	41.87
PCFA(ours)	60.37	76.08	48.43	9.50	64.85	4.74	**43.99**

In summary, based on the above comparative experiments, this study validates the feasibility of domain adaptation between these two datasets. Furthermore, for better visualization of our pixel-level feature classification performance, we selected pixel features during the training process for t-SNE [5], as shown in Fig. 8. It can be observed that as the network continues to learn, pixels from different classes gradually separate.

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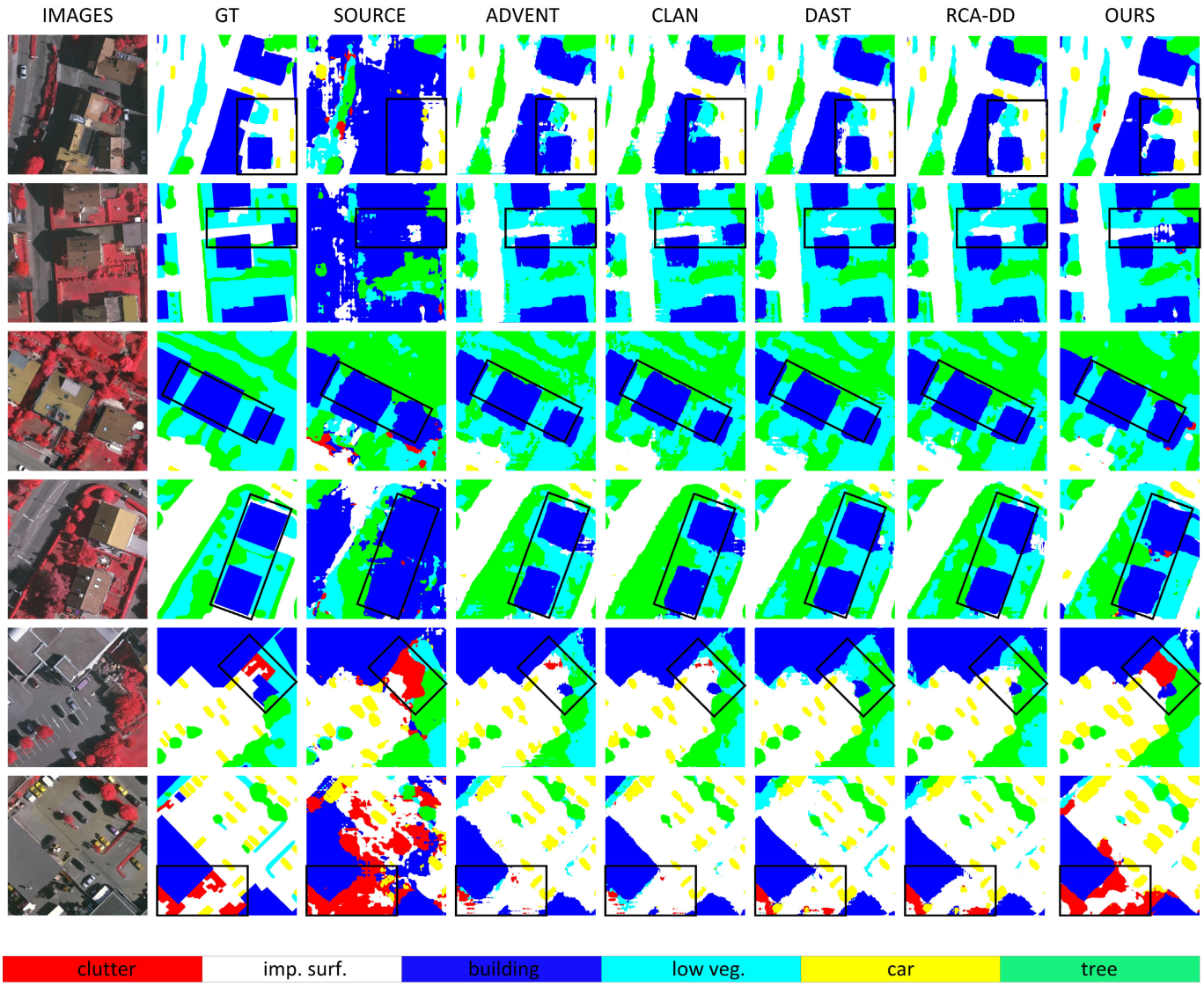


Fig. 6. Visualization results for the unsupervised DASS from the Potsdam IRRG to Vaihingen IRRG dataset. The black-bordered sections represent regions for ease of comparison. From left to right are the original image, ground-truth labels, source-only training, ADVENT [12], CLAN [18], DAST [35], RCA-DD [33], and ours (PCFA).

TABLE III

COMPARISON OF UNSUPERVISED DASS RESULTS OF THE POTSDAM IRRG TO VAIHINGEN IRRG DATASET. THE RESULTS ARE IOU (%). (∗) REPRESENTS EXPERIMENTAL RESULTS CITING OTHER ARTICLES. (†) DENOTES A REPLICATION EXPERIMENT USING THE DATASET DIVIDED IN THIS ARTICLE

Methods	Potsdam IRRG → Vaihingen IRRG						
	impervious surfaces	car	tree	low vegetation	building	clutter/background	mIou
†SOURCE ONLY	42.95	54.53	22.50	58.20	28.89	8.95	35.95
*ADVENT [12]	48.40	39.72	27.26	43.56	52.88	12.10	37.32
*CLAN [19]	50.95	39.17	25.50	46.39	53.27	13.21	38.08
*DAST [36]	57.40	41.59	34.32	47.73	57.90	13.07	42.00
*RCA-DD [34]	56.13	45.46	34.43	43.31	61.65	12.74	42.29
*DSM-A [53]	65.71	37.90	54.26	38.80	79.05	34.34	51.68
*CaGAN [56]	63.76	33.86	54.11	32.96	72.16	5.31	43.69
*DecoupleNet [55]	65.03	33.18	59.89	50.37	75.85	8.39	48.79
*MuGCDA [54]	69.12	35.28	61.57	51.31	75.42	11.34	50.67
PCFA(ours)	74.88	79.92	44.12	63.57	44.07	19.83	**54.40**

D. Ablation

1) PSM Storage Size: Following comparisons with other methods, to further validate the feasibility of our innovation, we conducted comprehensive ablation experiments regarding the setting of PSM module storage size. To make our parameter selection more reasonable, we designed multiple experimental groups with a wide range of coverage, and the specific results are summarized in Table V. From the table, it is

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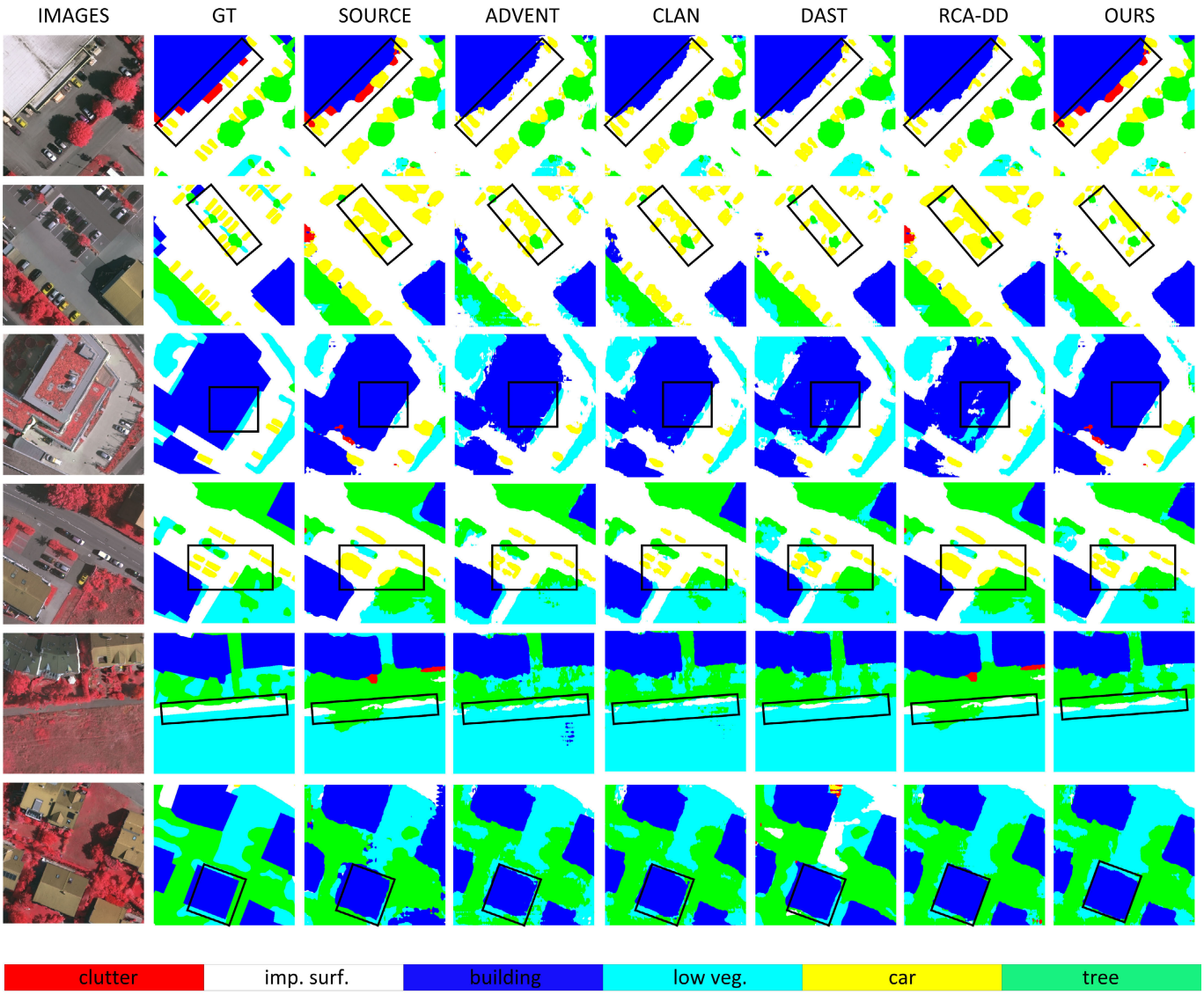


Fig. 7. Visualization results for the unsupervised DASS from the Potsdam RGB to Vaihingen IRRG. The black-bordered sections represent regions for ease of comparison. From left to right are the original image, ground-truth labels, source-only training, ADVENT [12], CLAN [18], DAST [35], RCA-DD [33], and ours (PCFA).

TABLE IV

COMPARISON OF UNSUPERVISED DASS RESULTS OF THE POTSDAM RGB TO VAIHINGEN IRRG DATASET. THE RESULTS ARE IOU (%). (∗) REPRESENTS EXPERIMENTAL RESULTS CITING OTHER ARTICLES. (†) DENOTES A REPLICATION EXPERIMENT USING THE DATASET DIVIDED IN THIS ARTICLE

Methods	Potsdam RGB → Vaihingen IRRG						
	impervious surfaces	car	tree	low vegetation	building	clutter/background	mIou
†SOURCE ONLY	40.35	55.36	17.42	60.21	30.13	10.31	35.63
*ADVENT [12]	52.14	17.29	53.84	20.44	47.53	2.26	32.25
*CLAN [19]	46.61	14.24	52.81	18.52	46.92	2.77	30.31
*DAST [36]	55.52	20.57	55.97	24.15	65.56	1.58	37.22
*RCA-DD [34]	51.27	17.35	53.52	17.71	50.13	0.34	31.72
*DSM-A [53]	60.75	38.21	51.81	22.78	70.10	31.94	45.93
*CaGAN [56]	58.46	32.19	57.55	20.15	68.21	9.73	41.04
*DecoupleNet [55]	52.99	33.19	61.03	17.31	61.52	4.30	38.39
*MuGCDA [54]	63.73	38.41	64.19	12.31	53.28	13.21	40.86
PCFA(ours)	75.20	93.45	33.36	89.57	83.39	62.38	**49.86**

observed that when the size is set to 0, meaning that the PSM module is not introduced, the metrics lag behind compared to when the module is introduced. Thus, our approach is effective. Upon examining the entire set of experiments, it is

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TABLE V

DIFFERENT SIZE OF PSM FOR UNSUPERVISED DASS. THE RESULTS OF VAIHINGEN IRRG → POTSDAM IRRG ARE IOU (%)

Size	Vaihingen IRRG → Potsdam IRRG						
	impervious surfaces	car	tree	low vegetation	building	clutter/background	mIou
0	59.53	72.80	52.16	44.93	65.49	12.48	51.23
5	61.98	69.96	58.04	48.53	62.14	11.21	51.98
10	63.71	71.69	56.26	47.86	64.62	8.59	52.12
50	63.99	72.61	57.26	46.12	67.05	7.93	52.49
100	62.84	70.49	57.25	44.52	67.26	8.62	51.83
250	62.88	70.91	55.32	45.7	63.00	17.63	**52.57**
500	63.56	71.65	56.73	47.37	60.01	9.88	51.53
1000	63.66	72.03	56.97	40.85	65.82	9.77	51.52

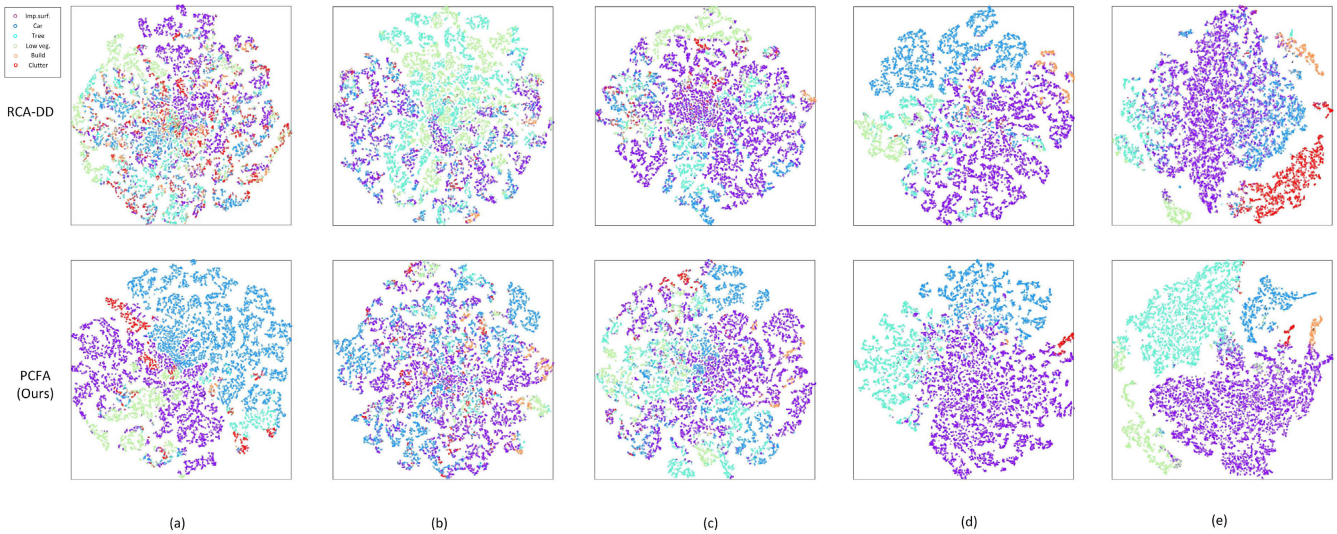


Fig. 8. We organized the pixel-level features of target-domain images during training and visualized them using t-SNE [5]. From left to right, the representation corresponds to the processes of the first, 500th, 2500th, 5000th, and 10 000th iterations. From the graph, it can be observed that with the increase in training epochs, pixels from different classes in the target domain are effectively separated, in comparison to RCA-DD [33]. (a)–(e) Results for the first, 500th, 2500th, 5000th, and 10000th iterations, respectively.

TABLE VI

ROLE PLAYED BY THE DIFFERENT MODULES IN PCFA (OURS) FOR UNSUPERVISED DASS. THE RESULTS OF VAIHINGEN IRRG → POTSDAM RGB ARE OA (%), MACC (%), AND MIOU (%)

Methods	PSM	EMD	PatchNCE	Vai IRRG → Pot RGB		
				OA	mAcc	mIou
Baseline	✗	✗	✗	51.92	48.78	34.82
	✗	✗	✔	66.68	57.34	42.53
PCFA (Ours)	✔	✗	✗	55.83	50.26	36.57
	✔	✗	✔	67.10	58.64	43.61
	✗	✔	✔	66.84	57.98	42.87
	✔	✔	✔	67.63	58.73	**43.99**

TABLE VII

ROLE PLAYED BY THE DIFFERENT MODULES IN PCFA (OURS) FOR UNSUPERVISED DASS. THE RESULTS OF POTSDAM IRRG → VAIHINGEN IRRG ARE OA (%), MACC (%), AND MIOU (%)

Methods	PSM	EMD	PatchNCE	Pot IRRG → Vai IRRG		
				OA	mAcc	mIou
Baseline	✗	✗	✗	63.92	64.32	42.04
	✗	✗	✔	76.45	74.31	52.14
PCFA (Ours)	✔	✗	✗	65.94	66.81	43.33
	✔	✗	✔	78.65	75.64	53.94
	✗	✔	✔	77.99	74.86	53.23
	✔	✔	✔	79.17	75.07	**54.40**

found that when the size is set to 250, the performance reaches its optimum, while values that are too large or too small result in a decline in metrics.

2) Interplay Between Components: For different types of image-transfer tasks, as observed in Table VI, the metrics are relatively lower compared to other tasks, indicating higher transfer difficulty. However, in comparison to the baseline, the modules in this article have a positive effect. When using EMD, the improvement is not substantial. However, with the introduction of the prototype-specific module (PSM), there is a significant enhancement. This suggests that for this type of supervisory mechanism, richer prototype features play a more critical role in network optimization. Metrics such as mIoU show an improvement of approximately 1%–2%. In the face of larger datasets and similar types of transfer tasks, the modules presented in this article are equally applicable, as depicted in Table VII. It is evident that the overall accuracy in the transfer tasks of this article consistently reaches the highest levels, and the performance improvement is closely related to each module. When transferring Potsdam RGB to Vaihingen IRRG in Table VIII, there is a significant drop in overall accuracy compared to the previous experiment due

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TABLE VIII

ROLE PLAYED BY THE DIFFERENT MODULES IN PCFA (OURS) FOR UNSUPERVISED DASS. THE RESULTS OF POTSDAM RGB → VAIHINGEN IRRG ARE OA (%), MACC (%), AND MIOU (%)

Methods	PSM	EMD	PatchNCE	Pot RGB → Vai IRRG		
				OA	mAcc	mIou
Baseline	✗	✗	✗	62.13	59.87	38.94
	✗	✗	✔	73.95	70.62	47.58
PCFA (Ours)	✔	✗	✗	63.96	60.58	39.87
	✔	✗	✔	74.61	71.81	49.17
	✗	✔	✔	74.22	71.43	48.95
	✔	✔	✔	75.03	72.89	**49.86**

TABLE IX

ROLE PLAYED BY THE DIFFERENT MODULES IN PCFA (OURS) FOR UNSUPERVISED DASS. THE RESULTS OF VAIHINGEN IRRG → POTSDAM IRRG ARE OA (%), MACC (%), AND MIOU (%)

Methods	PSM	EMD	PatchNCE	Vai IRRG → Pot IRRG		
				OA	mAcc	mIou
Baseline	✗	✗	✗	61.02	53.69	39.70
	✗	✗	✔	72.35	66.01	51.23
PCFA (Ours)	✔	✗	✗	63.23	55.29	41.17
	✔	✗	✔	74.01	67.64	52.57
	✗	✔	✔	73.68	66.46	51.47
	✔	✔	✔	74.89	67.57	**52.80**

TABLE X

NUMBER OF PARAMETERS, FLOPS, FORWARD PROPAGATION TIME, AND TEST TIME CORRESPONDING TO DIFFERENT METHODS. THE FORWARD PROPAGATION TIME IS MEASURED AS THE TIME REQUIRED FOR PROPAGATING ONE IMAGE IN THE POTSDAM DATASET WITH RGB CHANNELS

Methods	FLOPs	Parameters	Forward time	Test time
ADVENT [12]	190.21G	47.56M	1.29s	0.154s
CLAN [19]	197.82G	43.29M	1.03s	0.115s
DAST [36]	208.39G	50.28M	1.38s	0.101s
RCA-DD [34]	203.47G	45.17M	1.26s	0.180s
PCFA(ours)	210.46G	57.66M	1.45s	0.084s

to the change in data type. This is a common challenge in domain-adaptation tasks. However, the results of the ablation experiments suggest that the modules in this article are also well-suited for this type of transfer task, and each module has a positive effect. In the end, mIoU shows an improvement of 2.28% compared to the baseline. Through experiments in Table IX, it was observed that the overall performance is improved when only the PSM is utilized. However, when solely guided by the weighted prototype estimation, there is no significant improvement. It is only when these two components are combined that a substantial enhancement in overall accuracy is achieved. This is because the introduction of PSM brings richer prototype information to the network, enhancing its ability to memorize prototypes during training. Therefore, when optimizing the distribution of target-domain pixel features using the prototype's feature distribution, the effect is superior. These two components mutually reinforce each other, resulting in a final increase of 2.54% in OA, 0.86% in mAcc, and 1.53% in mIoU.

3) Comparison of Model Performance: We conducted a comprehensive evaluation of model characteristics, encompassing computational complexity and various pertinent metrics for several methods in Table X. The findings were systematically tabulated and subsequently subjected to a thorough analysis. The results clearly indicate that our approach is characterized by elevated computational complexity and a notably larger number of model parameters. This is an outcome of the inherently intricate architecture of our method, contributing to extended durations for both training phases.

V. CONCLUSION

This article primarily introduces a framework for unsupervised DASS. It leverages prototypes to establish a bridge between the source and target domains. Through the design of a prototype update module and the utilization of prototypes to guide network learning, we uncover the latent connections between these two domains. We enhance domain-adaptation performance on two distinct types of ISPRS datasets. Employing the principles of contrastive learning, we bring distances between pixels of the same class closer and those between pixels of different classes farther apart. Data-augmented images are used to construct positive and negative pairs required for the contrastive learning loss, providing the network with richer feature representations. To continually supervise the learning of target-domain features, we employ optimal transport theory, allowing the feature distribution of the prototypes in this article to interact with the feature distribution in the target domain, driving the target distribution toward a more representative prototype feature distribution. Experimental results demonstrate superior performance compared to some existing DASS methods in the remote-sensing domain.

In contrast to alternative methods, our approach exhibits a higher degree of structural intricacy, demanding considerable storage capacity for image features and incorporating prototypes, resulting in a discernible influence on computational resource usage. In the future, we envision further development of our approach involving model optimization. This entails the design of a streamlined domain-adaptation framework that strikes a balance between maintaining precision and enhancing operational efficiency, while concurrently reducing memory footprint. Meanwhile, we plan to further explore latent features among pixel representations to better adapt to semantic segmentation tasks in different complex domains.

REFERENCES

[1] H. Chen, H. Zhang, G. Yang, S. Li, and L. Zhang, "A mutual information domain adaptation network for remotely sensed semantic segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5537316.

[2] J. Hoffman, D. Wang, F. Yu, and T. Darrell, "FCNs in the wild: Pixel-level adversarial and constraint-based adaptation," 2016, *arXiv:1612.02649*.

[3] M. Gutmann and A. Hyvärinen, "Noise-contrastive estimation: A new estimation principle for unnormalized statistical models," in *Proc. 13th Int. Conf. Artif. Intell. Statist.*, 2010, pp. 297–304.

[4] F. I. Diakogiannis, F. Waldner, P. Caccetta, and C. Wu, "ResUNet-A: A deep learning framework for semantic segmentation of remotely sensed data," *ISPRS J. Photogramm. Remote Sens.*, vol. 162, pp. 94–114, Apr. 2020.

[5] L. Van der Maaten and G. Hinton, "Visualizing data using t-SNE," *J. Mach. Learn. Res.*, vol. 9, no. 11, 2008.

[6] Z. Jiang et al., "Prototypical contrast adaptation for domain adaptive semantic segmentation," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, Oct. 2022, pp. 36–54.

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[7] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Medical Image Computing and Computer-Assisted Intervention—MICCAI*. Munich, Germany: Springer, 2015, pp. 234–241.

[8] Y. Rubner, C. Tomasi, and L. J. Guibas, "The Earth Mover's distance as a metric for image retrieval," *Int. J. Comput. Vis.*, vol. 40, pp. 99–121, Nov. 2000.

[9] Q. Wang and T. Breckon, "Unsupervised domain adaptation via structured prediction based selective pseudo-labeling," in *Proc. AAAI*, 2020, vol. 34, no. 4, pp. 6243–6250.

[10] H. Noh, S. Hong, and B. Han, "Learning deconvolution network for semantic segmentation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1520–1528.

[11] V. Sandfort, K. Yan, P. J. Pickhardt, and R. M. Summers, "Data augmentation using generative adversarial networks (CycleGAN) to improve generalizability in CT segmentation tasks," *Sci. Rep.*, vol. 9, no. 1, p. 16884, Nov. 2019.

[12] T.-H. Vu, H. Jain, M. Bucher, M. Cord, and P. Pérez, "ADVENT: Adversarial entropy minimization for domain adaptation in semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2512–2521.

[13] X. Huang and S. Belongie, "Arbitrary style transfer in real-time with adaptive instance normalization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 1510–1519.

[14] Y. Zhang, P. David, and B. Gong, "Curriculum domain adaptation for semantic segmentation of urban scenes," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2039–2049.

[15] P. J. Soto Vega et al., "An unsupervised domain adaptation approach for change detection and its application to deforestation mapping in tropical biomes," *ISPRS J. Photogramm. Remote Sens.*, vol. 181, pp. 113–128, Nov. 2021.

[16] L. Zhang et al., "Unsupervised domain adaptation using robust class-wise matching," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 29, no. 5, pp. 1339–1349, May 2019.

[17] Y.-H. Tsai, W.-C. Hung, S. Schulter, K. Sohn, M.-H. Yang, and M. Chandraker, "Learning to adapt structured output space for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7472–7481.

[18] Y. Luo, L. Zheng, T. Guan, J. Yu, and Y. Yang, "Taking a closer look at domain shift: Category-level adversaries for semantics consistent domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2502–2511.

[19] L. Hoyer, D. Dai, and L. Van Gool, "DAFormer: Improving network architectures and training strategies for domain-adaptive semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 9914–9925.

[20] M. Kampffmeyer, A. B. Salberg, and R. Jenssen, "Semantic segmentation of small objects and modeling of uncertainty in urban remote sensing images using deep convolutional neural networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2016, pp. 1–9.

[21] T. Li, S. Roy, H. Zhou, H. Lu, and S. Lathuilière, "Contrast, stylize and adapt: Unsupervised contrastive learning framework for domain adaptive semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2023, pp. 4868–4878.

[22] B. Zhang, T. Chen, and B. Wang, "Curriculum-style local-to-global adaptation for cross-domain remote sensing image segmentation," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5611412.

[23] P. Russo, F. M. Carlucci, T. Tommasi, and B. Caputo, "From source to target and back: Symmetric bi-directional adaptive GAN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8099–8108.

[24] Y. Yang and S. Soatto, "FDA: Fourier domain adaptation for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 4084–4094.

[25] S. Sankaranarayanan, Y. Balaji, C. D. Castillo, and R. Chellappa, "Generate to adapt: Aligning domains using generative adversarial networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 8503–8512.

[26] R. A. Marsden, A. Bartler, M. Döbler, and B. Yang, "Contrastive learning and self-training for unsupervised domain adaptation in semantic segmentation," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jul. 2022, pp. 1–8.

[27] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2242–2251.

[28] T. Shen, D. Gong, W. Zhang, C. Shen, and T. Mei, "Regularizing proxies with multi-adversarial training for unsupervised domain-adaptive semantic segmentation," 2019, *arXiv:1907.12282*.

[29] T. Park, A. A. Efros, R. Zhang, and J. Y. Zhu, "Contrastive learning for unpaired image-to-image translation," in *Computer Vision—ECCV*. Glasgow, U.K.: Springer, 2020, pp. 319–345.

[30] R. Kemker, C. Salvaggio, and C. Kanan, "Algorithms for semantic segmentation of multispectral remote sensing imagery using deep learning," *ISPRS J. Photogramm. Remote Sens.*, vol. 145, pp. 60–77, Nov. 2018.

[31] J. Choi, T. Kim, and C. Kim, "Self-ensembling with GAN-based data augmentation for domain adaptation in semantic segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 6829–6839.

[32] Y.-C. Chen, Y.-Y. Lin, M.-H. Yang, and J.-B. Huang, "CrDoCo: Pixel-level domain transfer with cross-domain consistency," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 1791–1800.

[33] X. Chen, S. Pan, and Y. Chong, "Unsupervised domain adaptation for remote sensing image semantic segmentation using region and category adaptive domain discriminator," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4412913.

[34] Y. Li, T. Shi, Y. Zhang, W. Chen, Z. Wang, and H. Li, "Learning deep semantic segmentation network under multiple weakly-supervised constraints for cross-domain remote sensing image semantic segmentation," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 20–33, May 2021.

[35] F. Yu, M. Zhang, H. Dong, S. Hu, B. Dong, and L. Zhang, "DAST: Unsupervised domain adaptation in semantic segmentation based on discriminator attention and self-training," in *Proc. AAAI Conf. Artif. Intell.*, Feb. 2021, vol. 35, no. 12, pp. 10754–10762.

[36] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9726–9735.

[37] S. Ma et al., "Multicrop fusion strategy based on prototype assignment for remote sensing image scene classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5633212.

[38] S. Zhou et al., "DSM-assisted unsupervised domain adaptive network for semantic segmentation of remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5608216.

[39] Z. Xi et al., "A multilevel-guided curriculum domain adaptation approach to semantic segmentation for high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4404417.

[40] L. Yan, B. Fan, H. Liu, C. Huo, S. Xiang, and C. Pan, "Triplet adversarial domain adaptation for pixel-level classification of VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3558–3573, May 2020.

[41] S. Ma et al., "Automatic aug-aware contrastive proposal encoding for few-shot object detection of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5615211.

[42] D. Tuia, C. Persello, and L. Bruzzone, "Domain adaptation for the classification of remote sensing data: An overview of recent advances," *IEEE Geosci. Remote Sens. Mag.*, vol. 4, no. 2, pp. 41–57, Jun. 2016.

[43] T. Wu et al., "GINet: Graph interaction network for scene parsing," in *Computer Vision—ECCV*. Glasgow, U.K.: Springer, 2020, pp. 34–51.

[44] M. Y. Liu, T. Breuel, and J. Kautz, "Unsupervised image-to-image translation networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–9.

[45] F. Pan, I. Shin, F. Rameau, S. Lee, and I. S. Kweon, "Unsupervised intra-domain adaptation for semantic segmentation through self-supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 3763–3772.

[46] Z. Wang et al., "Differential treatment for stuff and things: A simple unsupervised domain adaptation method for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12632–12641.

[47] Y. Du, Z. Fu, Q. Liu, and Y. Wang, "Weakly supervised semantic segmentation by pixel-to-prototype contrast," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4310–4319.

[48] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

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[49] C. Li, B. Zhang, D. Hong, J. Yao, and J. Chanussot, "LRR-Net: An interpretable deep unfolding network for hyperspectral anomaly detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5513412.
[50] J. Yao, B. Zhang, C. Li, D. Hong, and J. Chanussot, "Extended vision transformer (ExViT) for land use and land cover classification: A multimodal deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5514415.
[51] D. Hong et al., "Cross-city matters: A multimodal remote sensing benchmark dataset for cross-city semantic segmentation using high-resolution domain adaptation networks," 2023, *arXiv:2309.16499*.
[52] X. Lai et al., "DecoupleNet: Decoupled network for domain adaptive semantic segmentation," in *Proc. Eur. Conf. Comput. Vis.* Cham, Switzerland: Springer, 2022, pp. 369–387.
[53] Q. Xu, X. Yuan, and C. Ouyang, "Class-aware domain adaptation for semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4500317.



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